MQTT教程 / MQTT Tutorial

什么是MQTT / What is MQTT?

MQTT (Message Queuing Telemetry Transport) 是一个轻量级的消息传输协议，基于发布/订阅模式设计。它最初由IBM开发，现已成为物联网（IoT）领域的标准协议之一。

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol designed based on the publish/subscribe pattern. Initially developed by IBM, it has become one of the standard protocols in the Internet of Things (IoT) field.

主要特点 / Main Features

优点 / Advantages

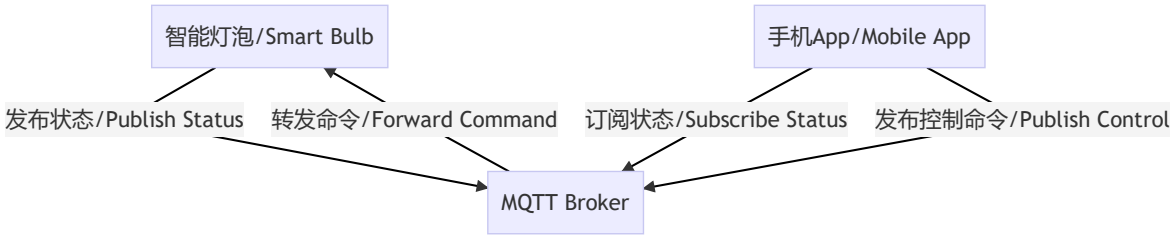
1. **轻量级 / Lightweight**
   - 协议简单，占用资源少 / Simple protocol, low resource consumption
   - 适合在带宽受限的网络中使用 / Suitable for bandwidth-constrained networks
   - 支持低功耗设备 / Supports low-power devices
2. **可靠性 / Reliability**
   - 提供三种服务质量级别(QoS) / Provides three Quality of Service (QoS) levels
   - 支持断线重连 / Supports connection recovery
   - 持久化会话 / Persistent sessions
3. **灵活性 / Flexibility**
   - 支持多种通信模式 / Supports multiple communication patterns
   - 主题可以灵活配置 / Flexible topic configuration
   - 支持一对多通信 / Supports one-to-many communication

缺点 / Disadvantages

1. **功能局限 / Functional Limitations**
   - 不适合大文件传输 / Not suitable for large file transfers
   - 没有内置安全机制 / No built-in security mechanisms
   - 需要额外的Broker服务器 / Requires additional Broker server
2. **依赖性 / Dependencies**
   - 依赖中心化的Broker / Depends on centralized Broker
   - 单点故障风险 / Single point of failure risk

典型应用场景 / Typical Use Cases

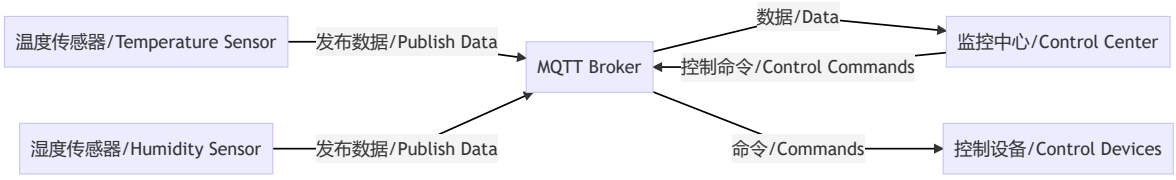
1. 智能家居 / Smart Home



示例主题 / Example Topics:

- home/livingroom/light/status
- home/livingroom/light/control

2. 工业监控 / Industrial Monitoring

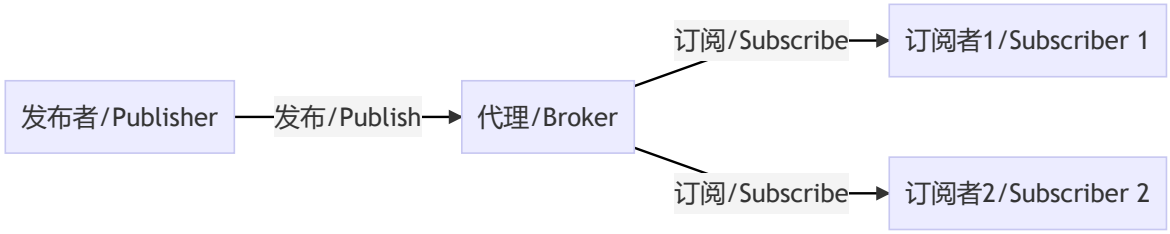


3. 车联网 / Connected Vehicles

- 车辆位置追踪 / Vehicle location tracking
- 远程诊断 / Remote diagnostics
- 实时状态监控 / Real-time status monitoring
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MQTT 基本流程 / Basic Concepts

发布-订阅模式 / Publish-Subscribe Pattern

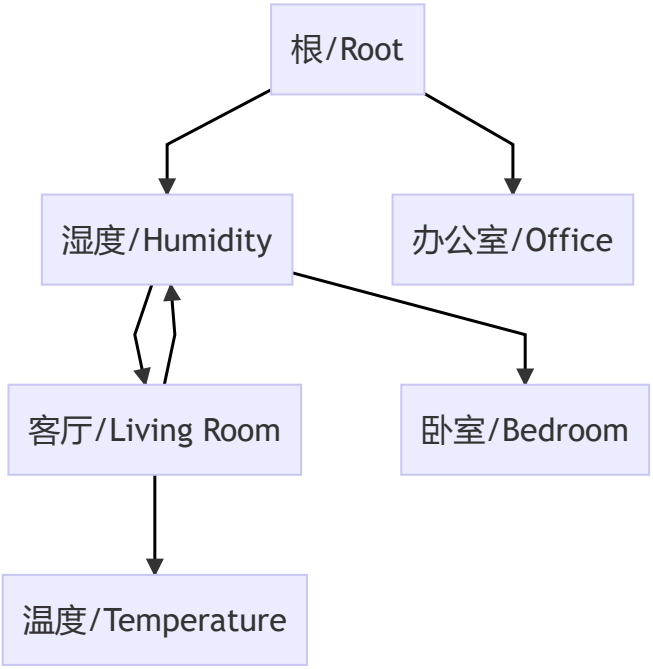


MQTT组件 / MQTT Components



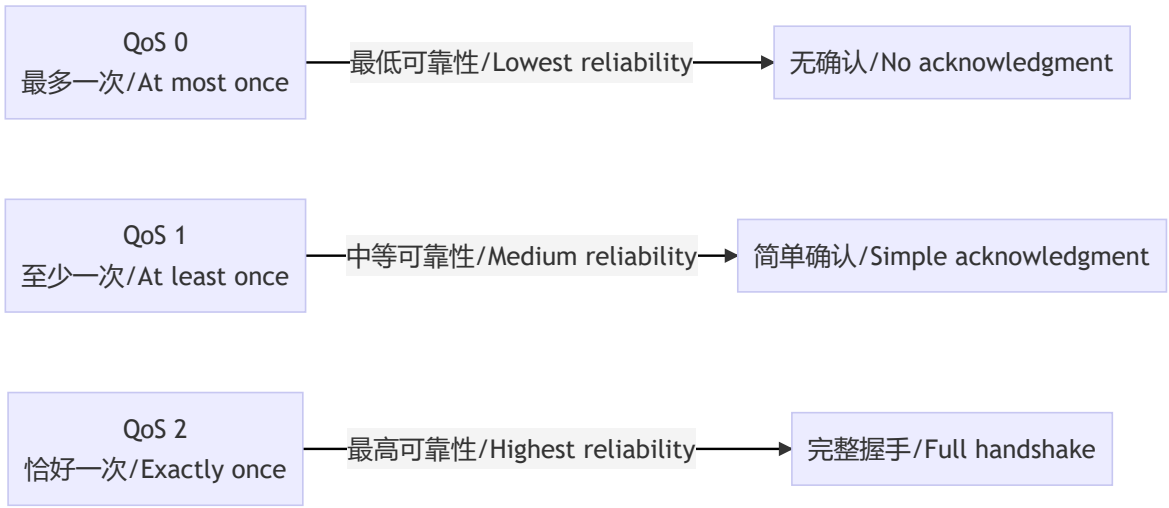
MQTT主题 / MQTT Topics

主题使用层级结构组织，使用斜杠(/)分隔各个层级。

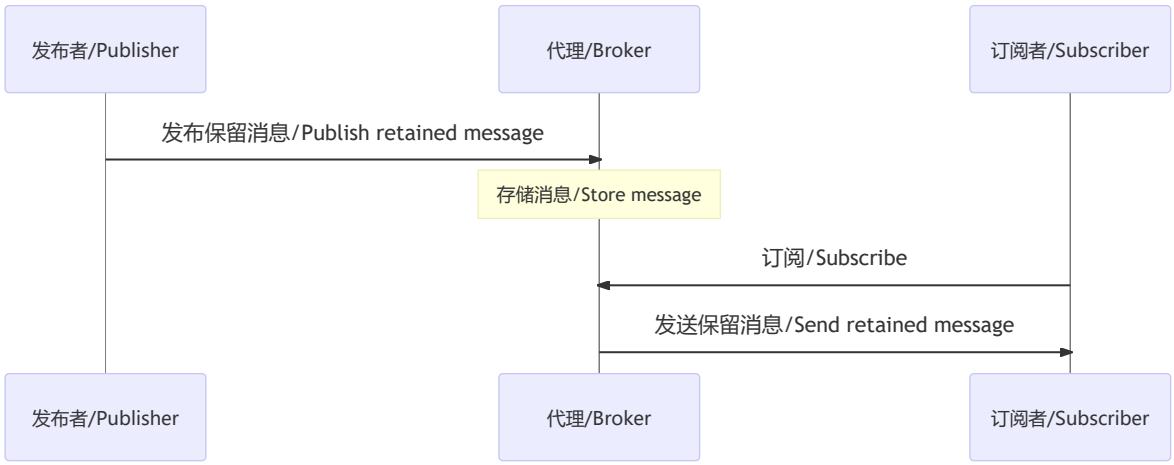


QoS级别 / QoS Levels

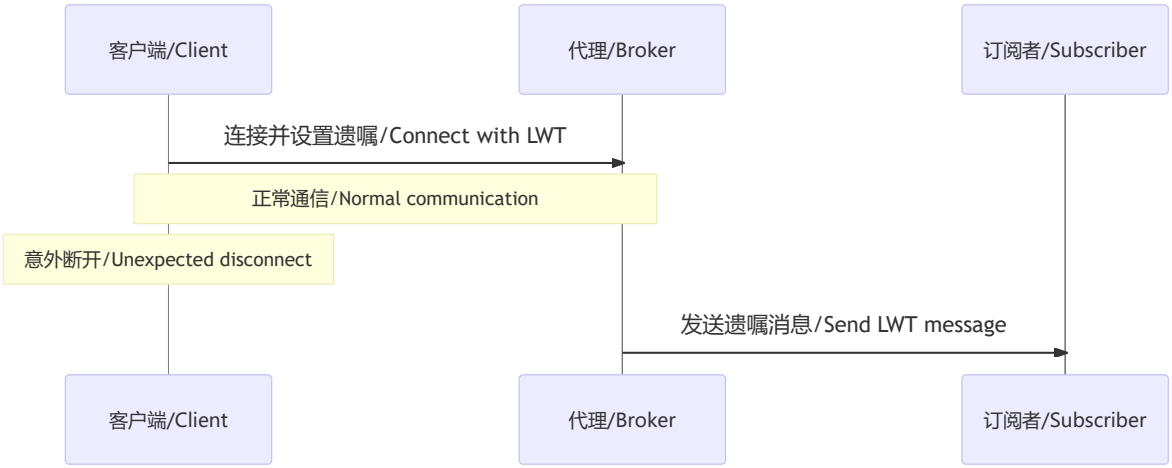
MQTT提供三种服务质量等级：



保留消息 / Retained Messages



遗嘱消息 / Last Will and Testament



代码示例 / Code Examples

> 代码示例请参考亚博智能K230教程 MQTT-发布者和 MQTT-订阅者章节